GRAPHICS LAB 05

From Phong Lighting to a Reflective Golden Stanford Bunny

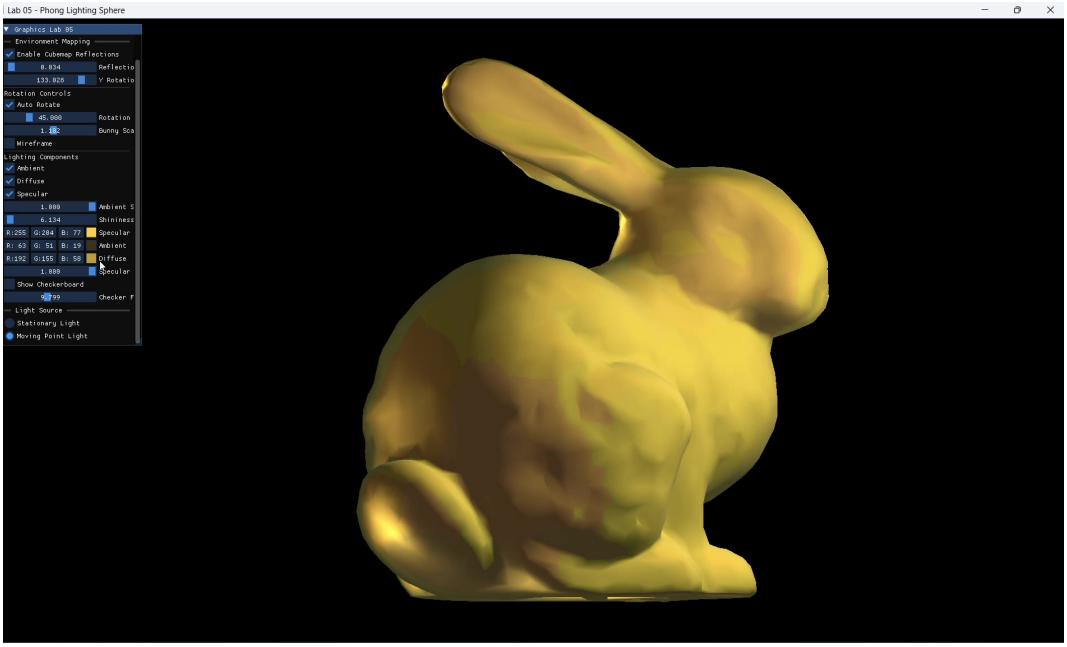
Technical progress report • C++ / OpenGL / GLSL / Dear ImGui • November 2026

Project objective

Explore how local lighting, surface geometry, procedural patterns, and environment mapping affect the appearance of 3D objects. The lab was developed incrementally: first a triangle, then a sphere, and finally a Stanford Bunny with an improved gold appearance.

Development timeline

1. Implement and investigate the Phong lighting components.
2. Apply lighting to a single triangle, the fundamental primitive of rasterized 3D meshes.
3. Move to a three-dimensional sphere and explore curved normals, highlights and a procedural checkerboard.
4. Load and render the Stanford Bunny mesh with gold-colored Phong material.
5. Introduce a six-face colored cubemap and adjustable environment-reflection blending.



Final-stage recording: golden Stanford Bunny with interactive lighting and cubemap controls.

1 Phong lighting: building the model

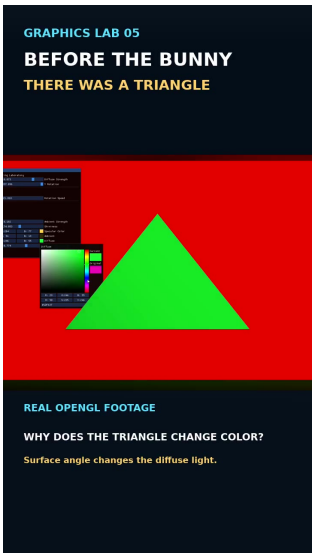
The lab began with the Phong reflection model, which approximates illumination as the sum of ambient, diffuse and specular contributions. Ambient provides baseline illumination; diffuse depends on the angle between the surface normal and light direction; specular produces view-dependent highlights controlled by a shininess exponent.

C = C_ambient + C_diffuse + C_specular

For a typical implementation: diffuse intensity is proportional to max(dot(N, L), 0), while Phong specular intensity uses pow(max(dot(R, V), 0), shininess). The vectors must be normalized and expressed in a consistent coordinate space.

2 Single triangle: the first lighting test

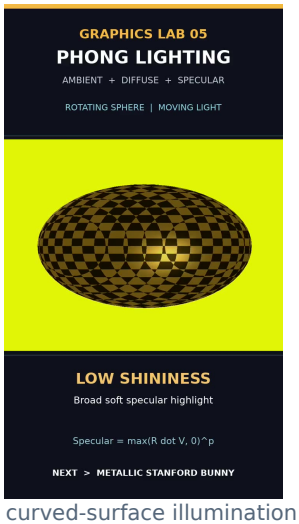
The first rendered geometry was a single triangle. Triangles are the basic primitives used to represent the surfaces of complex meshes in the rasterization pipeline. This made it possible to focus on the shader and the effect of light without introducing complicated geometry.



Triangle stage: testing the visual response to the lighting model.

3 Sphere: lighting on a curved surface

The experiment progressed to a 3D sphere. Unlike a flat triangle, the sphere has continuously varying surface normals, making changes in diffuse illumination and specular highlights easier to observe. A shininess control helped demonstrate the difference between broad and concentrated highlights.

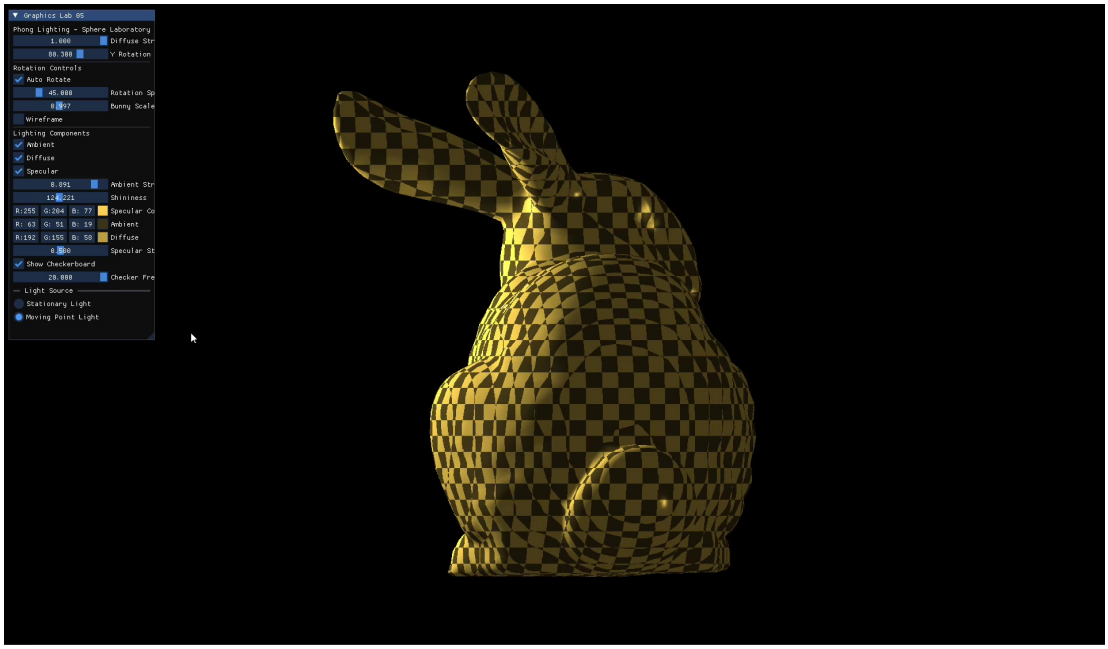


Sphere stage: inspecting curved-surface illumination and specular response.

4 Procedural checkerboard: inspecting the surface

A checkerboard was generated mathematically in the shader from object-space coordinates rather than loaded from an image. The pattern follows the geometry during rotation and makes lighting variation across the curved surface more visible. Because this is a 3D procedural pattern, it is not equivalent to UV image-texture mapping.

One possible implementation uses floor(localPosition * frequency) and alternates colors according to the parity of the summed cell coordinates. The checkerboard can affect ambient and diffuse contributions while specular highlights remain separately visible.



Recorded checkerboard bunny experiment: a procedural pattern under real-time lighting.

5 Stanford Bunny: moving from primitives to a mesh

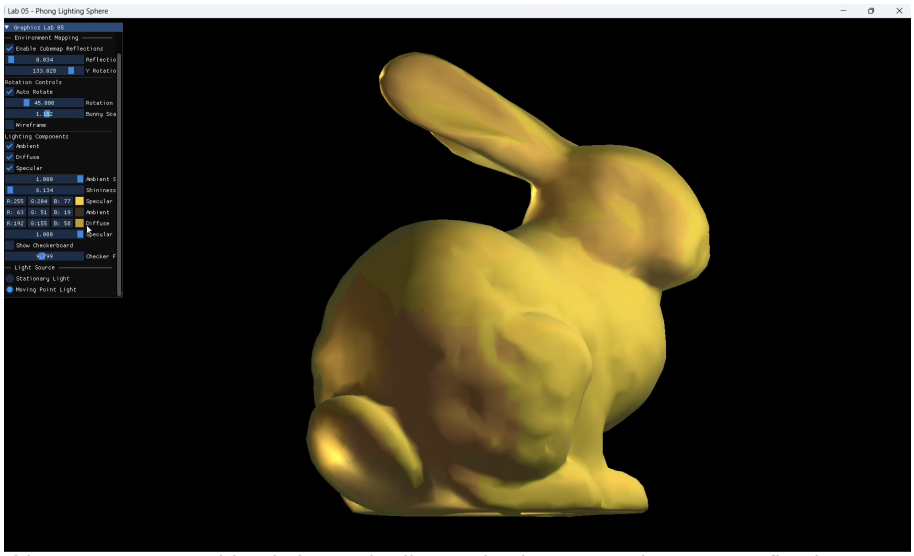
The lab then replaced the simple geometry with a Stanford Bunny OBJ mesh. Vertex positions, normals and triangle indices were used for indexed rendering. The bunny was rotated in real time, with lighting calculations responding to its surface orientation. The material was tuned toward a shiny gold appearance using color and shininess parameters.

A key implementation requirement is to transform normals consistently with the model rotation; camera position, fragment position and normal direction must be evaluated in compatible coordinate spaces.

6 Cubemapping: improving the gold appearance

A cubemap provides six directional texture faces (+X, -X, +Y, -Y, +Z and -Z). For each visible fragment, the shader computes a reflected view direction and samples the cubemap. The initial experiment used distinct solid colors for the six faces, enabling a clear demonstration of direction-dependent environment lookup without external image assets.

The reflected environment color was blended with the existing Phong result using an adjustable reflection strength. This produces an artistic approximation of reflective gold; it is not physically based rendering or ray tracing. The recorded final result achieved the desired gold appearance for this lab.



Final bunny stage: combined Phong shading and cubemap environment-reflection controls.

7 Interactive experiment controls

Control	Purpose
Ambient enable / strength	Toggle and adjust baseline illumination
Diffuse enable / strength	Toggle and adjust orientation-dependent lighting
Specular enable / strength	Toggle and adjust highlights
Shininess	Adjust concentration of specular highlights
Checkerboard enable / frequency	Compare solid and procedural surface patterns
Cubemap enable / reflection strength	Compare Phong-only and blended environment reflections

Conclusion and next challenge

Lab 05 progressed from the simplest triangle to a shaded, rotating golden bunny. Adjustable lighting components made each contribution observable, while checkerboard shading and cubemap sampling introduced two different ways to influence surface appearance. The golden result is considered satisfactory. The next lab challenge is image-based texture mapping, including UV coordinates and texture sampling in GLSL.

Portfolio: github.com/astavrop/ComputerGraphicsLabs | TikTok: @graphics.lab

Evidence note: Screenshots were extracted from available Lab 05 recordings. The report describes the demonstrated workflow and does not claim measured performance benchmarks.